

Machine Learning-Lecture 8

P1.py

```
C:\ml\L8>python p1.py
```

```
age sex cp trtbps chol fbs restecg thalachh exng oldpeak slp caa thall output
0  63  1  3  145 233  1    0   150  0  2.3  0  0  1  1
1  37  1  2  130 250  0    1   187  0  3.5  0  0  2  1
2  41  0  1  130 204  0    0   172  0  1.4  2  0  2  1
3  56  1  1  120 236  0    1   178  0  0.8  2  0  2  1
4  57  0  0  120 354  0    1   163  1  0.6  2  0  2  1
..  ... ..
298 57  0  0  140 241  0    1   123  1  0.2  1  0  3  0
299 45  1  3  110 264  0    1   132  0  1.2  1  0  3  0
300 68  1  0  144 193  1    1   141  0  3.4  1  2  3  0
301 57  1  0  130 131  0    1   115  1  1.2  1  1  3  0
302 57  0  1  130 236  0    0   174  0  0.0  1  1  2  0
[303 rows x 14 columns]
```

```
output
```

```
1 165
```

```
0 138
```

```
Name: count, dtype: int64
```

P2.py

```
C:\ml\L8>python p2.py
```

```
age sex cp trtbps chol fbs restecg thalachh exng oldpeak slp caa thall output
0  63  1  3  145 233  1    0   150  0  2.3  0  0  1  1
1  37  1  2  130 250  0    1   187  0  3.5  0  0  2  1
2  41  0  1  130 204  0    0   172  0  1.4  2  0  2  1
3  56  1  1  120 236  0    1   178  0  0.8  2  0  2  1
4  57  0  0  120 354  0    1   163  1  0.6  2  0  2  1
```

```

.. ... ..
298 57 0 0 140 241 0 1 123 1 0.2 1 0 3 0
299 45 1 3 110 264 0 1 132 0 1.2 1 0 3 0
300 68 1 0 144 193 1 1 141 0 3.4 1 2 3 0
301 57 1 0 130 131 0 1 115 1 1.2 1 1 3 0
302 57 0 1 130 236 0 0 174 0 0.0 1 1 2 0

```

[303 rows x 14 columns]

```

age      0
sex      0
cp       0
trtbps   0
chol     0
fbs      0
restecg  0
thalachh 0
exng     0
oldpeak  0
slp      0
caa      0
thall    0
output   0
dtype: int64

```

```

    age sex cp trtbps chol fbs restecg thalachh exng oldpeak slp caa thall
0   63  1  3  145 233  1    0   150   0  2.3  0  0  1
1   37  1  2  130 250  0    1   187   0  3.5  0  0  2
2   41  0  1  130 204  0    0   172   0  1.4  2  0  2
3   56  1  1  120 236  0    1   178   0  0.8  2  0  2
4   57  0  0  120 354  0    1   163   1  0.6  2  0  2
.. ... ..
298 57  0  0  140 241  0    1   123   1  0.2  1  0  3
299 45  1  3  110 264  0    1   132   0  1.2  1  0  3

```

```

300 68 1 0 144 193 1 1 141 0 3.4 1 2 3
301 57 1 0 130 131 0 1 115 1 1.2 1 1 3
302 57 0 1 130 236 0 0 174 0 0.0 1 1 2

```

[303 rows x 13 columns]

```

0 1
1 1
2 1
3 1
4 1

```

..

```

298 0
299 0
300 0
301 0
302 0

```

Name: output, Length: 303, dtype: int64

precision recall f1-score support

```

0 0.67 0.79 0.72 28
1 0.86 0.77 0.81 48

```

accuracy 0.78 76

macro avg 0.76 0.78 0.77 76

weighted avg 0.79 0.78 0.78 76

model saved

P3.py

C:\ml\L8>python p3.py

Age 23

Gender 1=male 0=female 1

chest pain type from 0 - 3 2

trtbps in mm hg 56

cholesterol level in mg/dl 33

fbs 67

restecg 23

thalachh 66

exng 34

oldpeak 6

slp 7

caa from 0-3 2

thall type 0/1/2/3 2

[0]

P4.py

C:\ml\L8>python p4.py

	Observations	Weather	Play
0	1	Sunny	No
1	2	Sunny	No
2	3	Overcast	Yes
3	4	Rainy	Yes
4	5	Rainy	Yes
5	6	Rainy	No
6	7	Overcast	Yes
7	8	Sunny	No
8	9	Sunny	Yes
9	10	Rainy	Yes
10	11	Sunny	Yes
11	12	Overcast	Yes
12	13	Overcast	Yes
13	14	Rainy	No

	Weather_Overcast	Weather_Rainy	Weather_Sunny
0	False	False	True
1	False	False	True
2	True	False	False
3	False	True	False
4	False	True	False
5	False	True	False
6	True	False	False
7	False	False	True
8	False	False	True
9	False	True	False
10	False	False	True
11	True	False	False
12	True	False	False
13	False	True	False

1. Overcast, 2. Rainy and 3. Sunny 2

['Yes']

['No' 'Yes']

pno = 37.75 % pyes = 62.25 %

P5.py

C:\ml\L8>python p5.py

	DN	Weather	Car	Result
0	1	sunny	working	go-out
1	2	rainy	broken	go-out
2	3	sunny	working	go-out
3	4	sunny	working	go-out
4	5	sunny	working	go-out
5	6	rainy	broken	stay-home
6	7	rainy	broken	stay-home

7 8 sunny working stay-home

8 9 sunny broken stay-home

9 10 rainy broken stay-home

Weather_rainy Weather_sunny Car_broken Car_working

0 False True False True

1 True False True False

2 False True False True

3 False True False True

4 False True False True

5 True False True False

6 True False True False

7 False True False True

8 False True True False

9 True False True False

weather - 1 rainy and 2 sunny 2

car - 1 broken and 2 working 2

['go-out']

[[0.94553707 0.05446293]]